

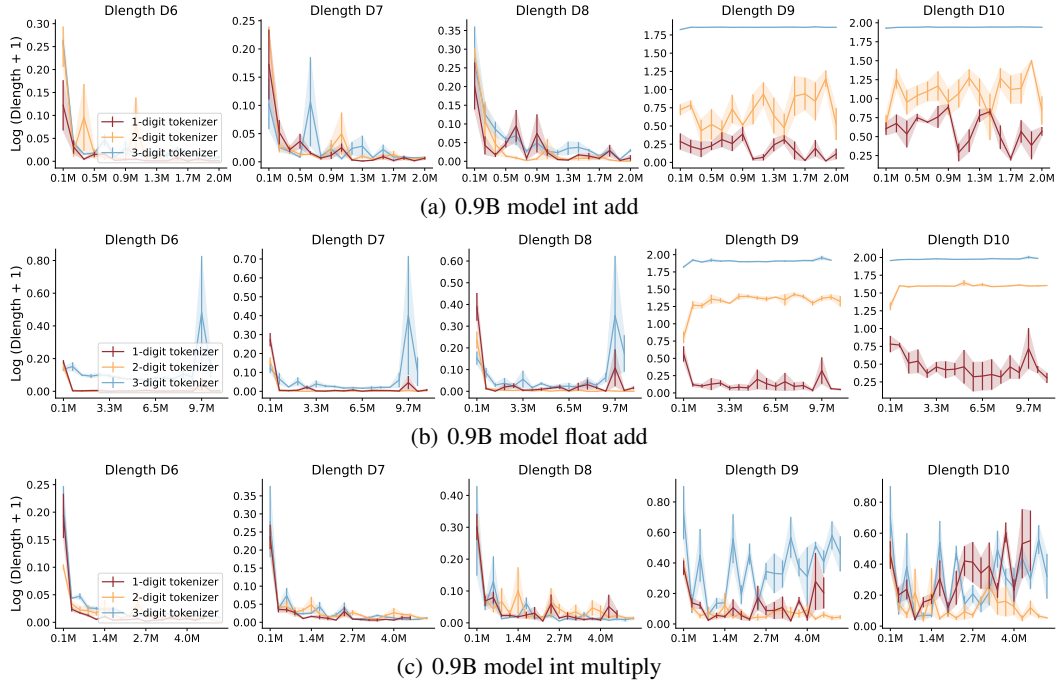


Figure 11: Accuracy of 0.9B models trained with 1-3 digit tokenizer on three tasks of integer addition, float addition and integer multiplication according to **dlength**. Here we report $\log_2(\text{dlength} + 1)$. X-axis is the number of seen training samples.

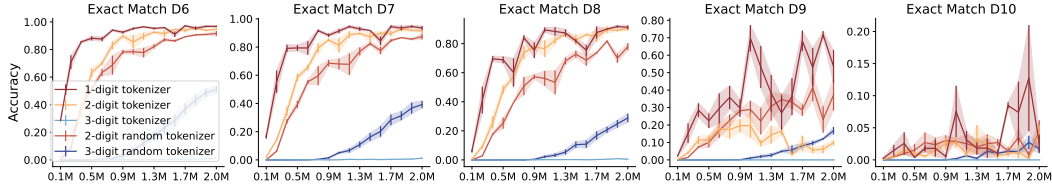


Figure 12: Accuracy of 0.9B models trained with 1- to 3-digit tokenizers and 2- to 3-digit random tokenizers on integer addition. Shadow shows the standard error. Dn means n digits. X-axis is the number of seen training samples.

Figure 12 shows the performance of 1- to 3-digit tokenizers alongside 2- to 3-digit random tokenizers, where n -digit random tokenizer means the one with maximal length n . In terms of in-domain generalization, the three-digit random tokenizer outperforms the three-digit standard tokenizer, while the two-digit random tokenizer shows a slight decline compared to its standard counterpart. We believe this is because the 0.9B model is capable of learning the two-digit tokenizer well, and the added perturbation from random tokenization acts as a form of regularization, introducing noise that slightly affects performance. The random tokenizers consistently outperform their standard counterparts in OOD generalization, indicating the regularization benefits in that aspect. In the case of the three-digit tokenizer, which is more challenging for a 0.9B model to learn, random tokenization generates smaller tokens, making the learning process easier and leading to improved in-domain performance. However, they **still fall short of the performance achieved by the one-digit tokenizer**.

A.4.3 PEs

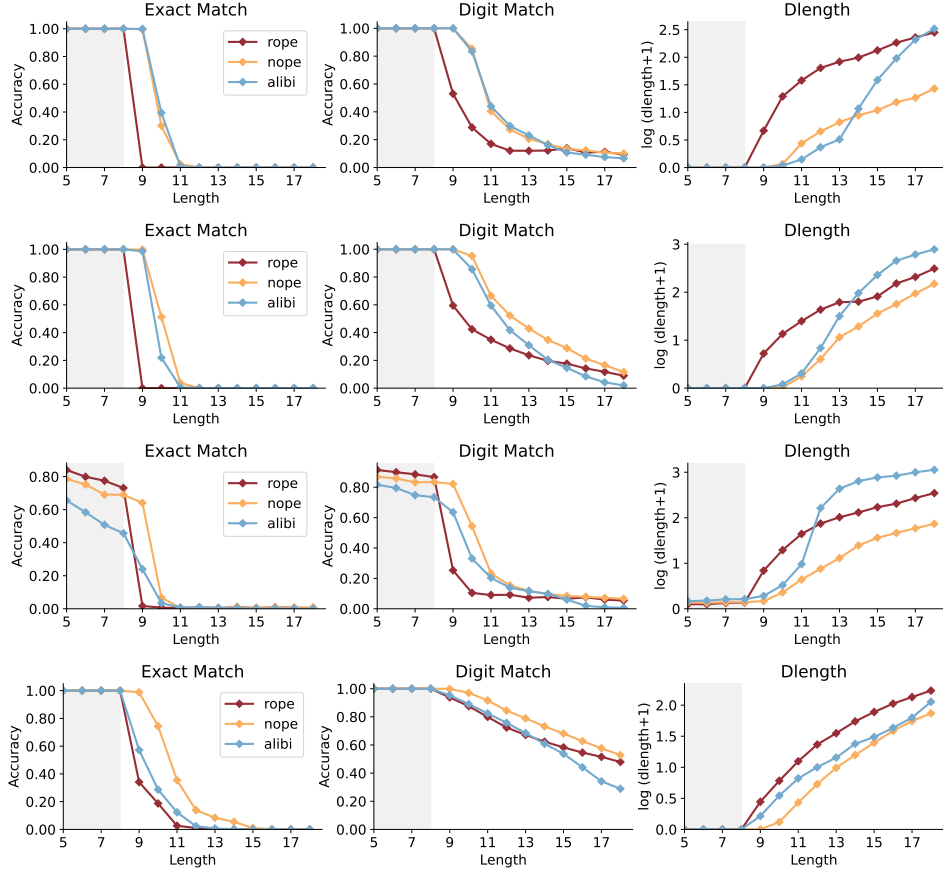


Figure 13: Exact match, digit match and dlength of 100M models trained with various PE, including RoPE, NoPE and Alibi. From top to bottom, the tasks are integer addition, float addition, fraction multiplication and scientific notation.

We show exact match, digit match and dlength of 100M models trained with various PE, including RoPE, NoPE and Alibi in Figure 13. We find NoPE and Alibi achieve better length generalization than RoPE, which is consistent with previous work like Zhou et al. (2024b).

To explain the mechanism of PEs, it is necessary to describe what the “generalization” is about. In most tasks, there is an intrinsic “length-agnostic” calculating rule, independent of the length of input numbers. For example, the addition rules: “align numbers by their least significant digits, add them digit by digit and carry over if the sum exceeds 9” is length-agnostic because it applies universally, regardless of the input length. However, during training on data with restricted length range (like 1 to 8), models may also learn *length-related* rules that fit the training data, such as combining normal addition rules with constraints like “the output length must range from 1 to 8”. Because these two rules are *indistinguishable*, prior knowledge should be added into the model as an inductive bias to help the model learn the “length-agnostic” rules expected in most practical settings (Abbe et al., 2024; Chen et al., 2024).

Table 12: RoPE performance with *standard error* from three repeated experiments. D_n means n digits where D8 is the longest in-domain length and D9 is the shortest out-of-domain length.

	Exact Match		Digit Match		Dlength	
	D8	D9	D8	D9	D8	D9
Int-add	1.00±0.00	0.00±0.00	1.00±0.00	0.45±0.02	0.00±0.00	1.07±0.02
Float-add	1.00±0.00	0.00±0.00	1.00±0.00	0.59±0.02	0.00±0.00	1.06±0.01
Frac-mul	0.70±0.01	0.01±0.00	0.85±0.01	0.22±0.02	0.18±0.02	1.45±0.08
Sci-add	1.00±0.00	0.23±0.08	1.00±0.00	0.92±0.01	0.00±0.00	0.66±0.11

According to our experiments, we find that (1) **RoPE encourages the model to rely on the length** of the input. The first evidence is that RoPE causes the model’s predictive performance to plummet dramatically just beyond the training boundary. We report the RoPE’s performance at the boundary of training length in Table 12 where D8 (digit 8) is the longest length in the training range while D9 (digit 9) is the shortest length out of the training range. In “*int-add*” task, the exact match drops from nearly 100% to 0% when moving from 8 to 9 digits, while “*dlength*” rises from 0 to 1.07 (Table 12). This indicates that the model has a significant probability of generating shorter results, avoiding the generation of more than 8-digit answers. At the same time, RoPE not only constrains the model’s output length but also affects the digit pairing. The performance of 100% for inputs of 8 digits indicates that the model performs calculations for each position unless it can successfully align the corresponding digits. However, when the model encounters 9-digit inputs, digit match drops significantly to 50%, suggesting a considerable probability of failing to align the digits. Similar results on the other three tasks suggest that it is a **task-agnostic** behavior. The only exception is the *digit match* of *scientific notation addition*. We discuss the results later.

(2) On the other hand, **length learning provided by RoPE appears to be a shortcut**. In cases where the model is extremely small or has been trained very little, we see the advantages of this “*shortcut*”. In Table 13, we train a 2-layer transformer (1.3M parameters) on integer addition using three different PEs on 1- to 8- digit integer addition or the 0.1B model with only 1M samples, we find RoPE shows the best in-domain performance. Experiments on the other three tasks are shown in Table 14 and Table 15, where the RoPE always surpasses others.

Table 13: 8-digit digit-match accuracy with small model or small dataset.

	1.3M Model	1M Samples
RoPE	0.091	0.97
NoPE	0.061	0.78
Alibi	0.056	0.23

As a possible explanation about why Alibi and NoPE achieve better length generalization, our experiments suggest that for length generalization in number tasks, the required inductive bias is to interpret the input as a sequence of digits while deliberately ignoring its overall length. RoPE, as a positional encoding that enables the model to quickly learn position-related information, may lead the model to adopt a length-dependent shortcut (Table 13), causing it to favor length-related rules. In contrast, both Alibi and NoPE diminish this reliance on position and length, encouraging the model to treat each unit’s operation as a step-by-step process, thereby achieving better length generalization.

Discuss about scientific addition The results in Table 12 reveal a clear trend where performance drops from 8-digit to 9-digit numbers, with one exception: the digit match score in the scientific notation addition task, which remains relatively high at 0.93 even for 9-digit numbers. We believe it is mainly because of the alignment mechanism between two scientific notations which differs from other representations. In other representations, numbers are aligned by **position** — integers from the most-left digit and the floats by the decimal point. However, in scientific notation, alignment depends on the difference in exponent values, which reduces RoPE’s reliance on position and mitigates length overfitting. Despite this, the effect of RoPE limiting output length remains apparent, as evidenced by the significant increase in the *dlength* score.